

A 10-bit 205-MS/s 1.0-mm² 90-nm CMOS Pipeline ADC for Flat Panel Display Applications

Seung-Chul Lee, *Member, IEEE*, Young-Deuk Jeon, *Member, IEEE*, Jong-Kee Kwon, *Member, IEEE*, and Jongdae Kim, *Senior Member, IEEE*

Abstract—This paper describes a 10-bit 205-MS/s pipeline analog-to-digital converter (ADC) for flat panel display applications with the techniques to alleviate the design limitations in the deep-submicron CMOS process. The switched source follower combined with a resistor-switch ladder eliminates the sampling switches and achieves high linearity for a large single-ended input signal. Multistage amplifiers adopting the complementary common-source topology increase the output swing range with lower transconductance variation and reduce the power consumption. The supply voltage for the analog blocks is provided by the low drop-out regulator for a high power-supply rejection ratio (PSRR) under the noisy operation environment. The pipeline stages of the ADC are optimized in the aspect of power consumption through the iterated calculation of the sampling capacitance and transconductance. The ADC occupies an active area of 1.0 mm² in a 90-nm CMOS process and achieves a 53-dB PSRR for a 100-MHz noise tone with the regulator and a 55.2-dB signal-to-noise-and-distortion ratio for a 30-MHz 1.0-V_{PP} single-ended input at 205 MS/s. The ADC core dissipates 40 mW from a 1.0-V nonregulated supply voltage.

Index Terms—Amplifiers, analog-to-digital converter (ADC), buffer circuits, CMOS analog integrated circuits, integrated circuit testing, low drop-out regulator, low power, low voltage, noise measurement, pipeline stage optimization, power-supply rejection ratio (PSRR), sample-and-hold (S/H) circuits, switched-capacitor circuits.

I. INTRODUCTION

WITH THE scaling down of CMOS technology, the increased unit-gain frequency and reduced device dimension promise the integration of multifunctional high-speed circuits with a smaller chip area and lower power consumption, whereas the thin gate oxide and short channel length, usable only in low supply voltages, cause several disadvantages in analog circuit design. The disadvantages are considered in the following four points of view: insufficient operating voltage margin, switch linearity degradation, leakage current, and low supply/substrate noise immunity.

The operating voltage margin is closely related to the signal swing and process variation. The circuit topologies sacrificing the voltage margin like cascode circuits are not only sensitive to process and supply variations but also have difficulty in achieving required specifications while maintaining a former signal swing lower than 1.0-V level supply voltages. On the other hand, the increased ratio of a signal swing to supply

voltages deteriorates the linearity of analog switches. The clock boosting schemes [1], [2] can enhance the switch linearity with a high gate voltage, which, however, is contrary to the trend of a lower gate breakdown voltage in the advanced CMOS process. Thus, circuit designers need to reduce the number of analog switches on the signal path or find out other methods to compensate for the lowered linearity.

In digital circuits, the static power dissipation by leakage current is a well-known problem of deep-submicron technology. However, in analog circuit design, the leakage current has different effects on the switched-capacitor circuits in that the leakage current changes the analog value memorized in a capacitor and affects the signal and noise transfer function by dramatically reducing the capacitor feedback factor over the low-frequency range. The last issue of substrate and supply noise immunities is a focal point in system-on-a-chip integration. How to measure supply noise in switched-capacitor circuits and how to minimize the noise effect in a specific functional block must be studied further as analog circuits become more sensitive to supply noise with the scaling down of supply voltage.

On the other hand, analog-to-digital converters (ADCs) are integrated in most of the recent systems because of their essential properties transferring real-world data in the analog domain to the computational world data in the digital domain. Moreover, as people want to analyze and process more real-world data, high-performance ADCs are demanded proportionally to the swelling information. Considering the portion of its power consumption and occupying area in an overall system, the integration of an ADC using scaled-down processes seems inevitable for a competitive system. The next-generation analog flat panel interface system demands ADC specifications that are unsuitable for the low supply voltages, such as a large single-ended input swing of a 1.0- V_{PP} level and noisy operation environment. Furthermore, since the system includes three high-speed ADCs with controllable offset and gain for independent RGB video signals, the small chip area and low power consumption are other important requirements in ADC design.

In this paper, we deal with three of above four issues, operating voltage margin, switch linearity, and low supply noise. The leakage current is not as critical a problem in wide-bandwidth applications, so we leave it to future studies. The pipeline ADC is optimized in the architectural level for low power consumption, and several circuit techniques are adopted to cope with the process limitations and handle the design specifications. In Sections II and III, the proposed pipeline ADC and pipeline stage optimization will be explained, respectively.

Manuscript received April 16, 2007; revised August 20, 2007.

The authors are with the Electronics and Telecommunications Research Institute, 305-350 Daejeon, Korea (e-mail: schlee@etri.re.kr).

Digital Object Identifier 10.1109/JSSC.2007.908760

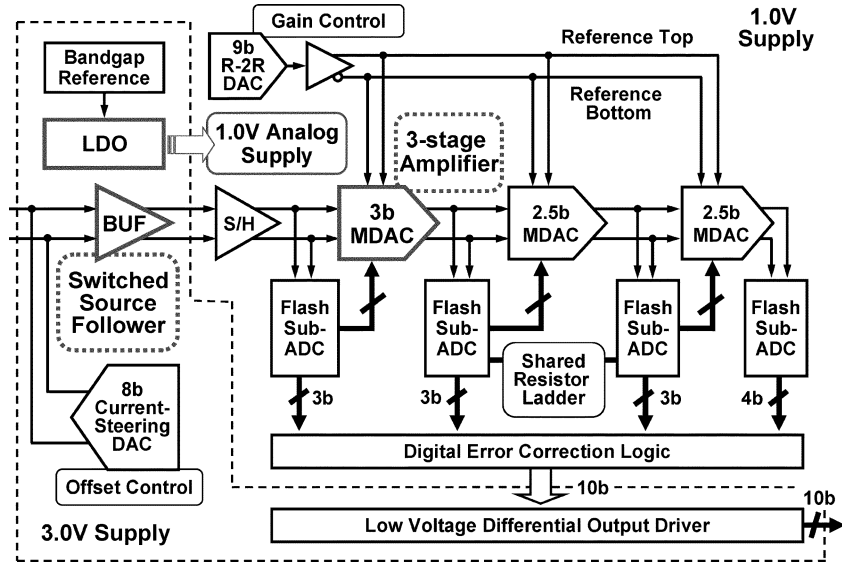


Fig. 1. Block diagram of the proposed ADC.

Then, Section IV will present the three circuit techniques of switched source follower, three-stage amplifier, and low drop-out (LDO) regulator, dealing with the disadvantages in the deep-submicron CMOS process. In the measurement section, Section V, the power-supply rejection ratio (PSRR) measurement method in switched-capacitor circuits will be proposed.

II. PROPOSED PIPELINE ADC ARCHITECTURE

The proposed ADC consists of an ADC core, two digital-to-analog converters (DACs) for gain and offset control, an LDO regulator, and an input buffer, as shown in Fig. 1. The flat-panel interface system furnishes both 3.0-V and 1.0-V supplies. Thus, the blocks directly connected to the input and output pads, such as the input buffer, the bandgap reference circuit, and the digital output driver, use 3.0-V I/O devices for better electrostatic protection and noise immunity, and analog and digital core blocks use 1.0-V supply for small chip area and low power consumption. The 1.0-V supply for the analog blocks is provided by the LDO regulator [3], [4] to isolate the analog supply from external supply noise with lower I/O supply voltages. When the regulator is off, the analog supply can be applied directly from off-chip.

The ADC core has multi-bit-per-stage pipeline architecture for optimizing power consumption and linearity, and the selection of the bit-per-stage will be described in the following section in detail. The mid-tread architecture is used for the symmetric design and layout resulting from its even characteristic. The first pipeline stage employs a 3 b multiplying DAC (MDAC), which means the first flash sub-ADC has eight comparators, and the gain of the MDAC is set to be four to correct digital errors caused by comparator offset. Consequently, the output swing of the MDAC is only the half of the input swing if there is no comparator offset. The rest of the stages have 2.5 b MDAC, which means the second and third flash sub-ADCs have six comparators, respectively. The second and third flash sub-ADCs, which operate at opposite clock phases, share a reference resistor ladder to reduce the chip area and power consumption [5].

The 9-bit R-2R single-ended DAC generates 0.25 V to 0.50 V, which is the voltage difference between the top and bottom reference voltages, which gives a 6 dB gain variation. The offset control is realized by the 8 b binary-coded current-steering DAC and the control range is ± 256 mV to change the video image brightness of each channel. The 10-bit digital output is converted into low-voltage overlapped differential signals, driving off-chip 100 Ω resistors, to facilitate high-speed digital capturing with low digital noise [6].

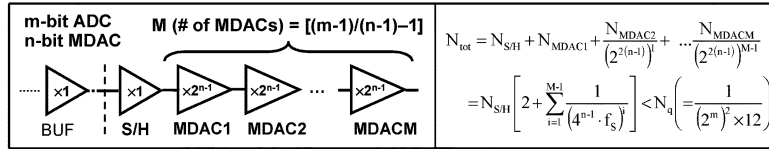
III. PIPELINE STAGE OPTIMIZATION

Determination of the bit-per-stage and scaling factor still remains as one of the unsettled study issues in pipeline ADC design because so many variables, such as sampling capacitance, parasitic capacitance, thermal noise, operating speed, and transconductance, are closely related to each other. The thermal noise of each stage is affected by both sampling capacitance and transconductance, and the transconductance increases according to the sampling capacitance and operating speed. Moreover, the parasitic capacitance of the amplifier, which is proportional to the transconductance, needs to be considered in the noise and transconductance calculation.

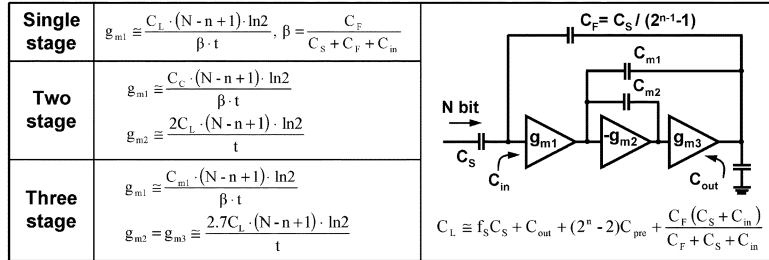
Although the previously published analyses [7]–[9] give the conceptual ideas for the stage optimization, it seems difficult to determine the practical values of sampling capacitors and transconductances with those approaches. For extracting the optimum values more exactly, as shown in Fig. 2, we propose the iteration method in which each variable is obtained by adopting several well-known equations repeatedly instead of figuring out the whole relationship at once with a single complex solution.

The first step of this method is to roughly estimate the minimum value of the sampling capacitors which make the total kT/C noise N_{tot} less than the quantization noise N_q for a given scaling factor f_s . In this step, the sampling capacitance of the sample-and-hold (S/H) circuit and first-stage MDAC is thought

Step 1: Determination of sampling capacitors (C_S) for a given scaling factor (f_s).



Step 2: Calculation of transconductances (g_{m1} , g_{m2} , and g_{m3}).



Step 3: Extraction of input and output parasitic capacitances (C_{in} and C_{out}).

Step 4: Iteration of Step 1 to 3 until variables converge on certain values.

Fig. 2. Iterated calculation method for the pipeline stage optimization.

to be the same value since the noise of each block $N_{S/H}$ and N_{MDAC1} gives the same effect on N_{tot} . The transconductance of each stage amplifier is not estimable, so the noise caused by active devices cannot be handled in the first turn. At this time, we just decide the number of amplifier stages in each block according to the linearity requirement.

Then, each stage transconductance can be calculated with the estimated sampling capacitance ignoring the input and output parasitic capacitance of amplifiers. The equations of Step 2 in Fig. 2 present the relation between the transconductance and other variables in single-, two-, and three-stage amplifiers, respectively. In the multistage amplifiers, which are usually employed in the front-stage MDACs and occupy a large portion of the overall ADC power consumption, the transconductance of the first-stage amplifier g_{m1} is inversely proportional to the feedback factor β , so the multi-bit-per-stage architecture uses more current at the first stage of the amplifier compared with the single-bit-per-stage architecture. However, at the second and third stages of the amplifier, the transconductances g_{m2} and g_{m3} are not related to β , but are related to the load capacitance C_L and settling accuracy $(N - n + 1)$. Thus, considering that g_{m1} is much smaller than g_{m2} and g_{m3} , the multi-bit-per-stage architecture has an advantage in power consumption if the capacitance of preamplifiers C_{pre} in the flash sub-ADCs is not dominant in C_L .

Now, the input and output parasitic capacitance of the amplifiers, C_{in} and C_{out} , can be calculated from the transconductances in the next step (Step 3). In the multistage amplifier, C_{in} is proportional to the first-stage transconductance g_{m1} , but C_{out} is proportional to the last-stage transconductance g_{m2} or g_{m3} . C_{out} also includes the capacitance of metal lines and switched-capacitor common-mode feedback circuits that are expected to be dependent on each stage transconductance. The constants between the transconductance and parasitic capacitance are extracted from the simulation results of each circuit structure.

Once the first values are determined, Steps 1) to 3) are iterated. During the iteration, the sampling capacitance and transconductance can be calculated more and more precisely using the previously ignored factors. If the transconductance and other variables converge on a certain value through the iteration, the current consumption is obtained from the ratios of the transconductance to drain current. For a given process and circuit topology, the ratio can be normalized within a certain range of the drain current because the transconductance is proportional to the square root of both the transistor size and drain current.

The simulation results of 10-bit 205-MS/s ADCs clearly illustrate that the multi-bit-per-stage architecture has an advantage for using the multistage amplifier in Fig. 3(a)–(c) compared with the single-bit-per-stage architecture because the multi-bit-per-stage architecture demands lower -3 dB bandwidth in closed-loop configuration, which reduces the power consumption of the second or third stage of the amplifier. Due to the large capacitance of preamplifiers in multi-bit-per-stage architectures, the 3-bit-per-stage ADC is expected to dissipate more current than the 2-bit-per-stage ADC does. However, between the 2-bit-per-stage ADCs and single-bit-per-stage ADC, this effect is less important than the -3 dB bandwidth requirement of the amplifier. Another interesting point is that, in multi-bit-per-stage architectures, the power consumption of the multistage amplifier of Fig. 3(b) is lower than that of the single-stage amplifier of Fig. 3(a) because, for the same drain current, the folded-cascode single-stage amplifier has a much smaller transconductance compared with the complementary common-source amplifier. In Fig. 3(d), we chose the lower taper factor of 0.5 (higher scaling factor) rather than the optimum point to reduce the power consumption of the input buffer driving the S/H circuit, where our design is 2-bit-per-stage architecture with 3-bit MDACs. The simulations for several different resolutions and clock frequencies are also performed, and the results show that, as the resolution and operating speed increase, the multi-bit-per-stage architecture is preferred.

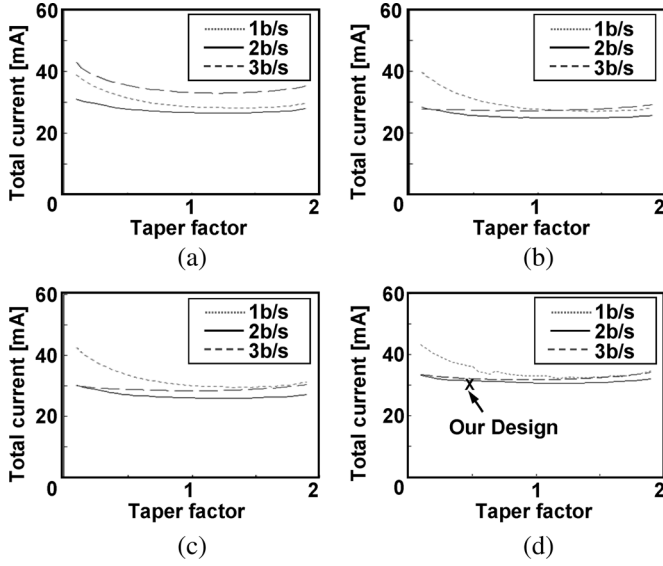


Fig. 3. Simulated total current of a 10-bit 205-MS/s ADC [(S/H + MDACs + flash sub-ADCs) versus taper factor (scaling factor = $1/(n-1)^{\text{taper factor}}$ [8]) (a) using only a single-stage amplifier, (b) using a two-stage amplifier in the first-stage MDAC, (c) using a three-stage amplifier in the first-stage MDAC, and (d) using the amplifiers satisfying the dc-gain requirement of each stage MDAC.

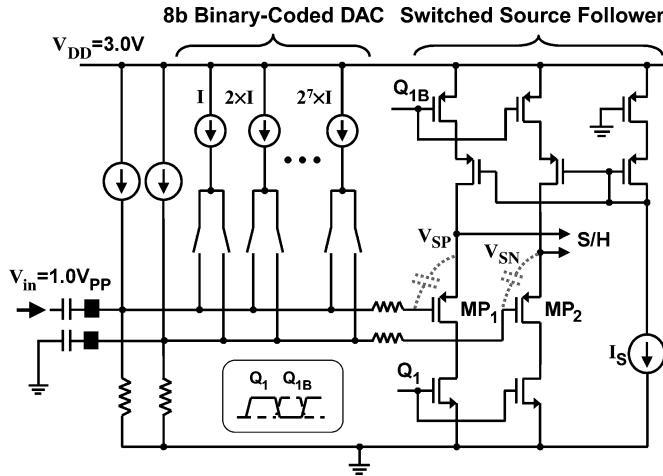


Fig. 4. Switched source follower and offset control DAC.

IV. LOW-VOLTAGE CIRCUIT DESIGN

A. Switched Source Follower and S/H Circuit

In this application, the input has a 1.0- V_{PP} single-ended swing and over 100-MHz bandwidth that make it difficult to achieve 10-bit level linearity of the sampling switch in an S/H circuit. Although clock bootstrapping improves the linearity, those techniques should be used carefully due to the low gate breakdown voltage of sub-0.1 μm CMOS processes. To guarantee high linearity without those techniques, as shown in Fig. 4, a switched source follower is used in front of the S/H circuit instead of the sampling switches and the conventional source follower, which has been used to drive the high-speed S/H circuit.

Although by this circuit the sampling switches are eliminated and high linearity can be achieved during the sampling phase,

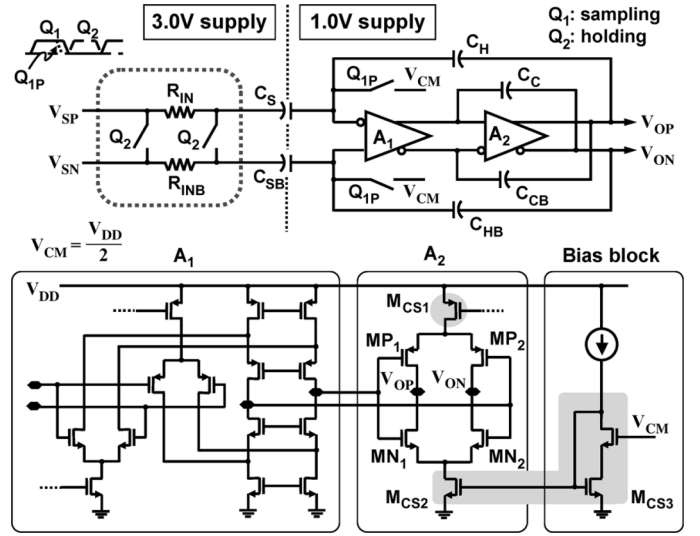


Fig. 5. S/H circuit employing a resistor-switch ladder.

the input signal coupling through the parasitic capacitors of the transistors MP_1 and MP_2 deteriorates the linearity of the sampled signal during the holding phase Q_2 of the S/H circuit. To reduce the coupling effect for high-frequency input signals, a resistor-switch ladder is adopted at the input stage of the S/H circuit in Fig. 5, which connects the bottom plates of the sampling capacitors C_S and C_{SB} during Q_2 . Typically, a single switch between the bottom plates of C_S and C_{SB} is used instead of the resistor-switch ladder, but the switch should be much larger than the sum of the switches in the resistor-switch ladder, thereby resulting in the larger input buffer that increases the coupling again.

The dc gain and bandwidth of amplifiers, in a low supply voltage, are more sensitive to supply and process variations because of the reduced gate-source and drain-source voltages, which demand an additional design margin. For a sufficiently high dc gain near 60 dB with a wide output swing, the S/H circuit employs a two-stage amplifier in Fig. 5. The complementary input pair of the second stage A_2 enhances the output swing with lower transconductance variation and reduces the power consumption compared with the conventional nMOS- or pMOS-only input pair topology in which the reduced transconductance seriously degrades the phase margin for a large-signal swing. The drain voltage of the bias transistor M_{CS3} is designed to have the same voltage as that of M_{CS2} , so the transistors can operate properly even in the deep linear region [10]. In this circuit, the maximum output swing is about $2 * (V_{thn} + |V_{thp}|)$, differentially. Therefore, the input devices MN_1 , MN_2 , MP_1 , and MP_2 with a higher threshold voltage can increase the output swing.

B. First-Stage MDAC With a Three-Stage Amplifier

The first-stage MDAC employs a three-stage amplifier with a reversed nested Miller compensation topology [11] in Fig. 6. The first stage of the three-stage amplifier has a folded-cascode architecture with an nMOS input pair for a higher feedback factor, and the second and third stages have the same complementary common-source topology as the second-stage amplifier

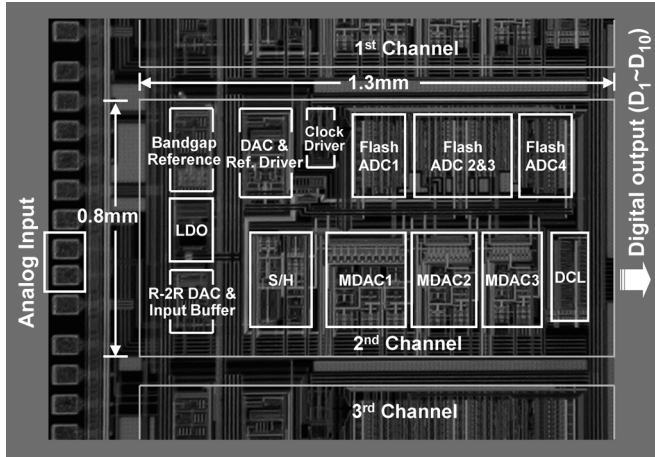
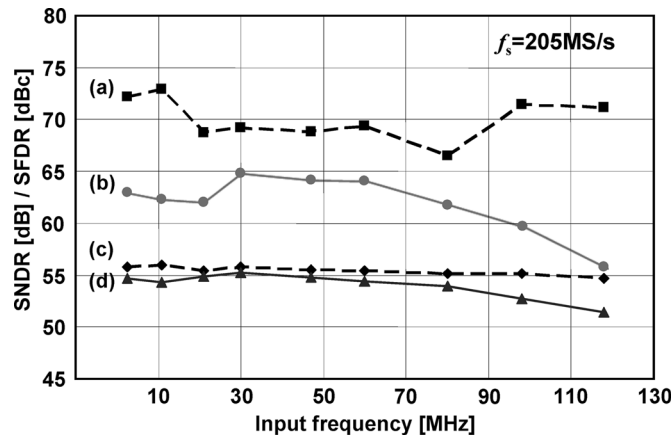


Fig. 9. Chip micrograph.

Fig. 10. Measured SNDR and SFDR versus input frequency. (a) SFDR for 1.0- V_{PP} differential input. (b) SFDR for 1.0- V_{PP} single-ended input. (c) SNDR for 1.0- V_{PP} differential input. (d) SNDR for 1.0- V_{PP} single-ended input.

V. MEASUREMENT RESULTS

The ADC of Fig. 9 is implemented in a 90-nm CMOS process, and its active area is 1.0 mm² (= 1.3 mm × 0.8 mm), including two DACs, an input buffer, a bandgap reference, and a regulator. Fig. 10 shows the measured dynamic performance with a varying input frequency at 205 MS/s, and the peak SNDR is 55.2 dB for a 1.0- V_{PP} differential input at low frequency. The SNDR of a 1.0- V_{PP} single-ended input is lower than that of the differential input because of the third-harmonic distortion caused by the front-end source follower, which is identified by comparing the SFDRs for the single-ended and differential input. An SFDR of more than 60 dBc is achieved at up to nearly 100-MHz frequency for a 1.0- V_{PP} single-ended input, and the input bandwidth could be improved by increasing the transistor size and current consumption of the source follower. For a 79-MHz 1.0- V_{PP} single-ended input, the SNDR is 53.9 dB and the SFDR is 61.8 dBc.

The ADC guarantees stable operations at several different clock frequencies (100 MS/s, 140 MS/s, 170 MS/s, and 205 MS/s) to support various video signal standards, and the SNDRs and SFDRs at each clock frequency are illustrated in Fig. 11(a) and (b). Fig. 11(c) and (d) shows the reconstructed

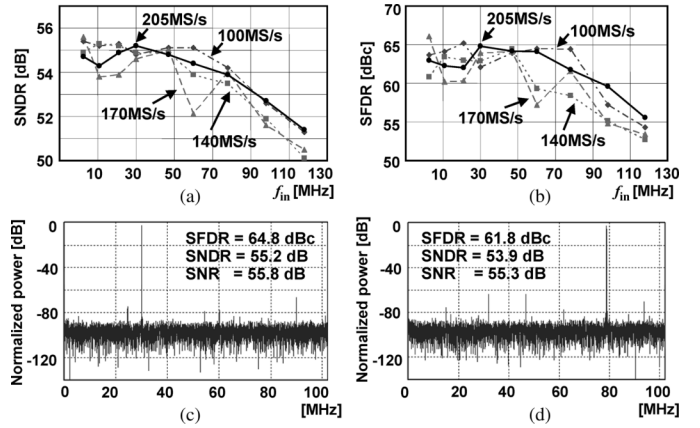
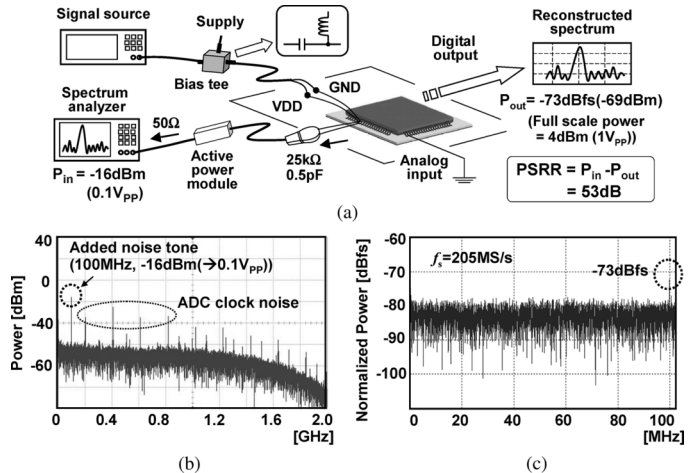
Fig. 11. Dynamic performances ($V_{in} = 1.0 V_{PP}$ single-ended). (a) SNDR versus input frequency. (b) SFDR versus input frequency. (c) $f_{in} = 30$ MHz, $f_s = 205$ MS/s. (d) $f_{in} = 79$ MHz, $f_s = 205$ MS/s.

Fig. 12. PSRR measurement. (a) Test setup. (b) Measured supply noise. (c) Reconstructed spectrum of the digital output.

spectrums with two different input frequencies, 30 MHz and 79 MHz, at 205 MS/s. Every measurement is performed for a 1.0- V_{PP} single-ended input.

With the scaling down of supply voltages, the PSRR limits the dynamic performance of ADCs more and more. However, the PSRR for high-frequency noise is rarely measured in switched-capacitor circuits because the supply fluctuation includes not only the external noise but also the noise generated by its own clock operation. Furthermore, the test board has a certain amount of capacitance between the supply and ground, so only a small amount of the applied high-frequency noise is transferred to the board.

To measure the PSRR for the external high-frequency supply noise exactly, we apply the noise tone through the bias tee and measure the noise directly from the supply and ground pins of the test chip by the wide-bandwidth high-impedance active probe and spectrum analyzer, as illustrated in Fig. 12(a). Then, the noise amplitude is calculated from the noise power in the frequency domain, and the noise tone of the reconstructed spectrum is measured from the ADC digital output, shown in Fig. 12(b) and (c), respectively. In this setup, the analog input

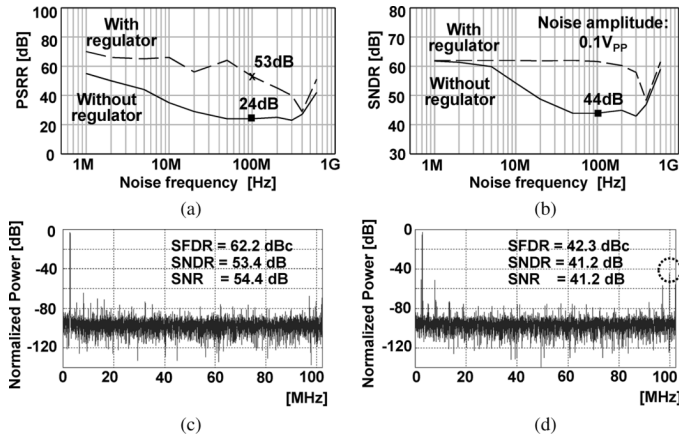


Fig. 13. Relation between PSRR and SNDR. (a) Measured PSRR. (b) Maximum achievable SNDR of an ideal 10-bit ADC. (c) Reconstructed spectrum with regulator ($f_{in} = 2.5$ MHz, $f_s = 205$ MS/s, $f_{noise} = 100$ MHz, $V_{noise} = 100$ mV_{PP}). (d) Reconstructed spectrum without regulator ($f_{in} = 2.5$ MHz, $f_s = 205$ MS/s, $f_{noise} = 100$ MHz, $V_{noise} = 100$ mV_{PP}).

pin of the ADC should be connected to ac ground to prevent the noise from directly affecting the input stage.

When the regulator operates, the PSRR is improved by more than 20 dB for up to 200-MHz supply noise compared to when the regulator does not operate. In Fig. 13(a), the PSRRs with and without the regulator are 53 and 24 dB, respectively, for 100-MHz noise. Fig. 13(b) plots the maximum achievable SNDR of an ideal 10-bit ADC calculated from the measured PSRR when 100-mV_{PP} noise is added to the analog supply. The maximum SNDR without regulator is expected to be only around 44 dB for 100-MHz noise frequency. Fig. 13(c) and (d) illustrate the measured reconstructed spectrum with and without the regulator, respectively, for a 100-MHz 100-mV_{PP} noise tone. In this test board, the SNDR without any external noise is only 53.5 dB because all power-coupling capacitors are eliminated from the board for applying the external power supply noise to the chip efficiently. The supply noise pattern of digital systems is also measured, which shows that subharmonic tones of the clock, $f_{clk}/2, f_{clk}/3, 2 \times (f_{clk}/3), f_{clk}/4, 3 \times (f_{clk}/4) \dots$, generated from the transition of digital circuits appear as supply noise.

The PSRR for the substrate noise should be measured separately. In this test setup, the substrate is also bounding due to the external noise, so the results include the effects by supply and substrate noise. If there exists a certain amount of internal noise and the effect of bonding wire is not so critical, it would give the similar degradation of ADC performances with that for the same amount of external noise. Therefore, a system engineer could estimate the performance degradation of an ADC before system integration from the noise pattern of a certain system (whether it is internal or external) and the PSRR of the ADC.

The measured differential nonlinearity and integral nonlinearity are ± 0.5 LSB and ± 0.5 LSB, respectively, for a 1.0- V_{PP} differential signal in Fig. 14(a). The INL is slightly bent in a symmetrical fashion by the distortion of the front-end switched source follower. Fig. 14(b) obtained from a 1.0- V_{PP} single-ended input illustrates this distortion more clearly. When the input reduces to a 600-mV_{PP} differential signal, the INL decreases to less than 0.4 LSB.

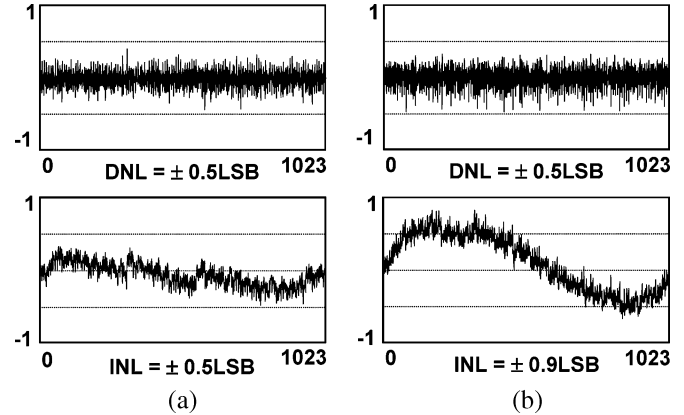


Fig. 14. Measured DNL and INL for (a) 1.0- V_{PP} differential input. (b) 1.0- V_{PP} single-ended input.

Resolution	10b	
Sampling speed	205MS/s	
Input range	1.0V _{PP} (Single-ended)	
Power consumption (@1.0V supply)	With regulator	Without regulator
	111mW	61mW
SNDR	$f_{in}=30$ MHz	55.2dB
	$f_{in}=79$ MHz	53.9dB
Active area (per channel)	1.0mm ² (=1.3mm × 0.8mm)	
Technology	90nm 1P6M CMOS	

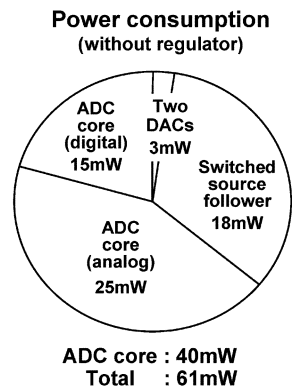


Fig. 15. Performance summary of the prototype ADC.

The performance of the ADC is summarized in Fig. 15. The total power consumption of a single-channel ADC is 111 mW for a 30-MHz sinusoidal input signal at 205 MS/s with the regulator operated at a 3.0-V supply. The ADC core dissipates 40 mW for a 1.0-V supply, so, when the regulator is not operated and the analog supply is applied directly from off-chip, the total power consumption is about 61 mW, including the input buffer and two DACs power of 21 mW.

VI. CONCLUSION

This paper attempted to alleviate the limitations in the deep-submicron CMOS process, insufficient operating voltage margin, switch linearity degradation, and low supply/substrate noise immunity, using three design techniques. The switched source follower combined with a resist-switch ladder releases the switch linearity degradation in low supply voltages. Multi-stage amplifiers adopting the complementary common-source topology achieves a high gain and sufficient operating voltage margin for a large single-ended input signal. The supply voltage for the analog blocks is provided by the LDO regulator for higher PSRRs. By measuring the PSRR for high-frequency noise, we can estimate the performance degradation of ADCs before system integration. On the other hand, the optimization method presented in this paper calculates the power consumption of pipeline ADCs more exactly, which gives the optimum sampling capacitance and transconductance of each pipeline stage before circuit design.

REFERENCES

- [1] T. Cho and P. R. Gray, "10-b, 20-Msample/s, 35-mW pipeline A/D converter," *IEEE J. Solid-State Circuits*, vol. 30, no. 3, pp. 166–172, Mar. 1995.
- [2] A. Abo and P. R. Gray, "A 1.5-V 10-bit 14 MS/s CMOS pipeline analog-to-digital converter," in *Proc. Symp. VLSI Circuits*, Jun. 1998, pp. 166–169.
- [3] G. A. Rincon-Mora and P. E. Allen, "A low-voltage, low-quiescent current, low drop-out regulator," *IEEE J. Solid-State Circuits*, vol. 33, no. 1, pp. 36–44, Jan. 1998.
- [4] K. N. Leung and P. K. T. Mok, "A capacitor-free CMOS low-dropout regulator with damping-factor-control frequency compensation," *IEEE J. Solid-State Circuits*, vol. 38, no. 10, pp. 1691–1702, Oct. 2003.
- [5] Y.-D. Jeon, S.-C. Lee, K.-D. Kim, J.-K. Kwon, and J. Kim, "A 5-mW 0.26-mm² 10-bit 20-MS/s pipelined CMOS ADC with multi-stage amplifier sharing technique," in *Proc. Eur. Solid-State Circuits Conf.*, Montreux, Switzerland, 2006, pp. 544–547.
- [6] S.-C. Lee *et al.*, "A 10-bit 400-MS/s 160-mW 0.13- μ m CMOS dual-channel pipeline ADC without channel mismatch calibration," *IEEE J. Solid-State Circuits*, vol. 41, no. 7, pp. 1596–1605, Jul. 2006.
- [7] S. H. Lewis, "Optimizing the stage resolution in pipelined, multistage, analog-to-digital converter for video-rate applications," *IEEE Trans. Circuits Syst II, Analog Digit. Signal Process.*, vol. 39, no. 8, pp. 516–523, Aug. 1992.
- [8] D. W. Cline and P. R. Gray, "A power optimized 13-b 5-Msamples/s pipelined analog-to-digital converter in 1.2 μ m CMOS," *IEEE J. Solid-State Circuits*, vol. 31, no. 3, pp. 294–303, Mar. 1996.
- [9] Y. Chiu, P. R. Gray, and B. Nikolic, "A 14-b 12-MS/s CMOS pipeline ADC with over 100-dB SFDR," *IEEE J. Solid-State Circuits*, vol. 39, no. 12, pp. 2139–2151, Dec. 2004.
- [10] K. Gulati and H. S. Lee, "A high-swing CMOS telescopic operational amplifier," *IEEE J. Solid-State Circuits*, vol. 33, no. 12, pp. 2010–2019, Dec. 1998.
- [11] R. Eschauzier and J. Huijsing, *Frequency Compensation Techniques for Low-Power Operational Amplifiers*. Boston, MA: Kluwer, 1995.



Seung-Chul Lee (S'98–M'00) was born in Seoul, Korea, in 1973. He received the B.S. and M.S. degrees in electronic engineering from Sogang University, Seoul, Korea, in 1998 and 2000, respectively.

Since July 2000, he has been with the Electronics and Telecommunications Research Institute (ETRI), a nonprofile government-funded research organization, Daejeon, Korea, where he is currently a Senior Researcher. His work focuses mainly on mixed-mode integrated circuit design, and his current interest is in communication systems based on high-resolution

high-speed CMOS data converters and low-voltage low-power CMOS analog circuit design.



circuit design.

Young-Deuk Jeon (S'98–M'00) was born in Busan, Korea, in 1975. He received the B.S. and M.S. degrees in electronic engineering from Sogang University, Seoul, Korea, in 1998 and 2000, respectively.

Since September 2005, he has been with the Electronics and Telecommunications Research Institute (ETRI), Daejeon, Korea, where he is currently involved with the design of the analog front-end circuit for image signal processing. His research has been focused on high-resolution low-power CMOS data converters and low-voltage CMOS analog



Jong-Kee Kwon (M'94) received the B.S. and M.S. degrees in electronics engineering from the Yeungnam University, Korea, in 1981 and 1983, respectively, and the Ph.D. degree in electrical engineering from the Korea Advanced Institute of Science and Technology (KAIST), Seoul, in 2004.

Since 1984, he has been with Electronics and Telecommunications Research Institute (ETRI), Daejeon, Korea, where he has been involved mainly with the development of analog integrated circuits (ICs). As the Technical Manager of the Mixed-Signal

Product Engineering Department of ETRI, he currently is involved in the development of high-speed, high-performance mixed-signal ICs including low-voltage/low-power analog functions, amplifiers, high-speed/high-resolution data converters, PLLs/VCOs, and analog interfaces for sensor or MEMS systems.



Jongdae Kim (S'89–M'94–SM'02) received the B.S. and M.S. degrees in electronics engineering from Kyungpook National University, Taegu, Korea, in 1982 and 1984, respectively, and the Ph.D. degree in electrical and computer engineering from the University of New Mexico, Albuquerque, in 1994.

From 1984 to 1989, he was with the Electronics and Telecommunications Research Institute (ETRI), Daejeon, Korea, where he worked on silicon-based device design and process integration of EEPROM and CMOS. In 1994 he rejoined ETRI. His current research

interests include power integrated circuits for FED, PDP, and OLED driving ICs, micro dc–dc converters, and circuit design and IPs based on nono-technology. He is now a Director of the IT-NT Research Group at ETRI. He has published over 80 technical papers in international journals and conference proceedings.

Dr. Kim is a Senior Member of the IEEE Electron Device Society and the Institute of Electronics Engineering of Korea.